

SYDNEY TECHNICAL HIGH SCHOOL



HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 3

JUNE 2015

Mathematics Extension 2

General Instructions

- Working time - 90 minutes
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown in questions 6 to 9
- Full marks may not be awarded for careless work or illegible writing
- Start each question on a new page
- All answers are to be in the writing booklet provided
- Marks for each question are indicated on the question
- A table of standard integrals is provided at the back of this paper

Total marks - 55

Section 1 - 5 marks

Attempt Questions 1 – 5.
Allow about 10 minutes for this section.

Section 2 - 50 marks

Attempt Questions 6 – 9.
Allow about 80 minutes for this section.

Name : _____

Teacher : _____

Section 1 (5 marks)

Attempt Questions 1 – 5

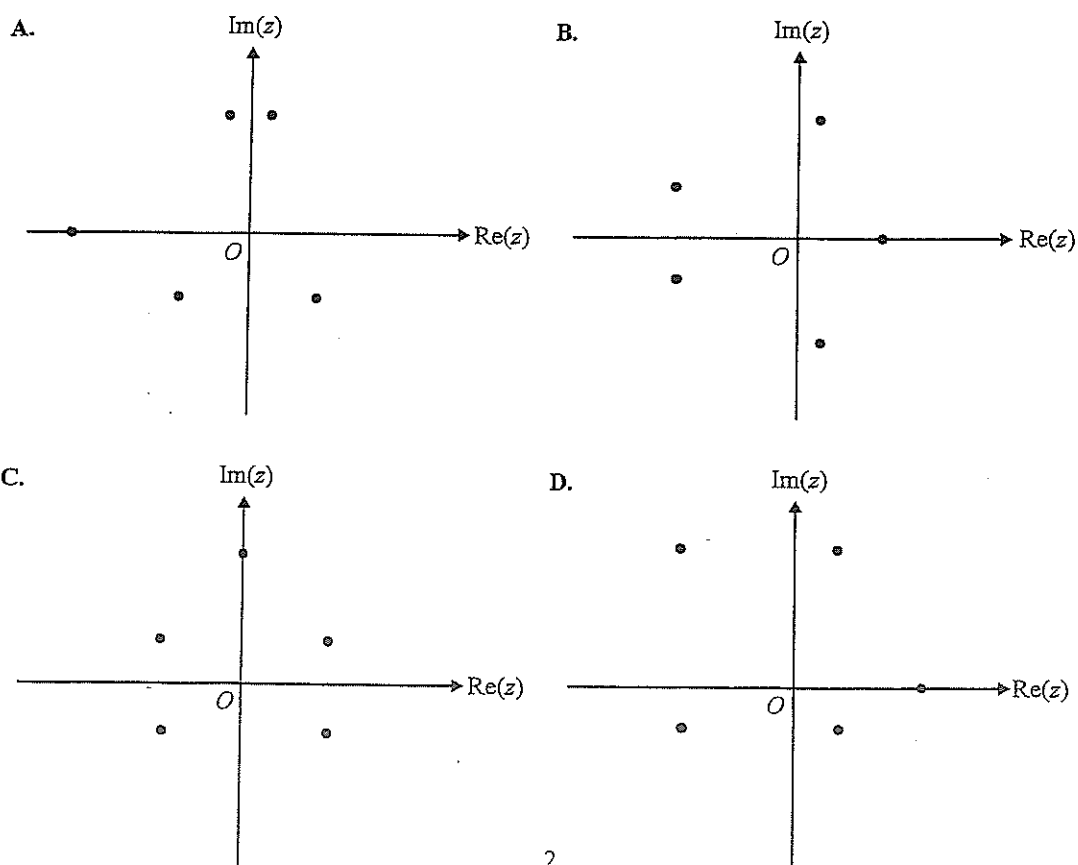
Use the multiple-choice answer sheet in your answer booklet for Questions 1 – 5.
Do not remove the multiple-choice answer sheet from your answer booklet.

1. The polynomial $P(x)$ of degree 4 has real coefficients.
 $P(x)$ has roots α, β, γ and δ and it is known that $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = -10$.

Which of the following must be true ?

- (A) $P(x)$ has all its roots real.
- (B) $P(x)$ has one real and three imaginary roots.
- (C) $P(x)$ has two real and two imaginary roots.
- (D) $P(x)$ has at least two imaginary roots.

2. Which one of the following diagrams could represent the location of the roots of $z^5 + z^2 - z + c = 0$ in the complex plane, given that c is real ?



3. With a suitable substitution, $\int_0^{\frac{\pi}{3}} \cos^2 x \sin^3 x \, dx$ can be expressed as

(A) $\int_{0.5}^1 u^2 - u^4 \, du$

(B) $\int_1^{0.5} u^2 - u^4 \, du$

(C) $\int_0^{\frac{\pi}{3}} u^2 - u^4 \, du$

(D) $-\int_0^{\frac{\sqrt{3}}{2}} u^2 - u^4 \, du$

4. Which one of the following is a primitive function of $\frac{6}{\sqrt{1-4x^2}}$?

(A) $3 \sin^{-1}(2x)$

(B) $6 \sin^{-1}(2x)$

(C) $12 \sin^{-1}\left(\frac{x}{2}\right)$

(D) $3 \sin^{-1}\left(\frac{x}{2}\right)$

5. $\int_0^a \left(\sin^2\left(\frac{3x}{2}\right) - \cos^2\left(\frac{3x}{2}\right) \right) dx$ is equal to

(A) $-\frac{4}{3} \sin\left(\frac{3a}{4}\right)$

(B) $-\frac{1}{3} \sin(3a)$

(C) $\frac{1}{3} \sin(3a)$

(D) $\frac{1}{3} (1 - \sin(3a))$

Section 2 (50 marks)

Attempt Questions 6 – 9

Start each question on a new page

Question 6 (12 marks)

- (a) Given the polynomial $P(x) = 3x^4 - 14x^3 + 12x^2 + 24x - 32$ 3
has a triple root, solve the equation $P(x) = 0$.
- (b) Find $\int \frac{dx}{x^2 - 6x + 11}$. 3
- (c) Use the substitution $u = -x$ to evaluate $\int_{-1}^1 \frac{dx}{e^x + 1}$. 3
- (d) Using the trigonometric identity $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, or otherwise, 3
solve the polynomial equation $8x^3 - 6x + 1 = 0$,
giving your answers correct to 3 decimal places.

Question 7 (13 marks) (Start a new page in your answer booklet)

(a) Find $\int \frac{dx}{x^2+6x-7}$. 3

(b) Use the substitution $x = 3 \sin \theta$ to evaluate $\int_0^{\frac{3}{\sqrt{2}}} \sqrt{9-x^2} \, dx$. 4

(c) The polynomial $P(x) = x^3 - 5x^2 + 8x + b$, where b is a constant, has a factor in the form $(x - k)^2$.

(i) Show that the possible values of k are $\frac{4}{3}$ and 2. 3

(ii) For $k = 2$, find the value of b and hence fully factorise $P(x)$. 3

Question 8 (13 marks) (Start a new page in your answer booklet)

(a) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{5+3 \cos x-4 \sin x}$ using the substitution $t = \tan \frac{x}{2}$. 4

(b) If α, β and γ are the roots of the equation $x^3 + 4x^2 + 3x - 3 = 0$
find the polynomial equation whose roots are

(i) $\frac{\alpha}{2}, \frac{\beta}{2}$ and $\frac{\gamma}{2}$. 2

(ii) $\alpha\beta - 1, \alpha\gamma - 1$ and $\beta\gamma - 1$ 3

(c) (i) If $I_n = \int_1^e (1 - \ln x)^n dx, n \geq 0$ 2

show that $I_n = -1 + n I_{n-1}, n \geq 1$.

(ii) Hence, or otherwise, evaluate $\int_1^e (1 - \ln x)^3 dx$ 2

Question 9 (12 marks) (Start a new page in your answer booklet)

- (a) (i) Use the substitution $x = u^2$, $u > 0$, to show that 4

$$\int_4^{16} \frac{\sqrt{x}}{x-1} dx = 4 + 2\ln 3 - \ln 5.$$

- (ii) Hence use integration by parts to evaluate 2

$$\int_4^{16} \frac{\ln(x-1)}{\sqrt{x}} dx$$

- (b) (i) Solve the equation $z^5 + 1 = 0$ over the complex field, 2
giving the complex roots in the form $r(\cos\theta + i\sin\theta)$.

- (ii) If α is the complex root of $z^5 + 1 = 0$ with smallest positive argument, 2
show that the other complex roots can be expressed as $-\alpha^2$, α^3 and $-\alpha^4$.

- (iii) If α is the complex root of $z^5 + 1 = 0$ with smallest positive argument, 2
form the quadratic equation with roots $\alpha - \alpha^4$ and $\alpha^3 - \alpha^2$,
giving your answer in the form $ax^2 + bx + c = 0$.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

EXTENSION 2 SOLUTIONS - JUNE 2015

1. D

2. B

3. A

4. A

5. B

6. a)
$$P(x) = 3x^4 - 14x^3 + 12x^2 + 24x - 32$$
$$P'(x) = 12x^3 - 42x^2 + 24x + 24$$
$$P''(x) = 36x^2 - 84x + 24$$

Solving
$$36x^2 - 84x + 24 = 0$$
$$3x^2 - 8x + 2 = 0$$
$$(3x-1)(x-2) = 0$$
$$x = \frac{1}{3}, 2$$

$$P'(\frac{1}{3}) \neq 0, \quad P'(2) = 0$$

$\therefore$ triple root is $x=2$

$$\therefore 2 + 2 + 2 + \alpha = \frac{14}{3}$$
$$\alpha = -1\frac{1}{3}$$

$\therefore$ solutions $2, 2, 2, -1\frac{1}{3}$

b)
$$\int \frac{dx}{x^2 - 6x + 11}$$

$$= \int \frac{dx}{(x-3)^2 + 2}$$

$$= \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x-3}{\sqrt{2}}\right) + C$$

$$c) \int_{-1}^1 \frac{dx}{e^x + 1}$$

$$u = -x$$

$$du = -dx$$

$$= \int_1^{-1} \frac{-du}{e^{-u} + 1}$$

$$= \int_{-1}^1 \frac{du}{\frac{1}{e^u} + 1}$$

$$= \int_{-1}^1 \frac{e^u du}{1 + e^u}$$

$$= \ln(1 + e^u) \Big|_{-1}^1$$

$$= \ln(1 + e) - \ln(1 + e^{-1})$$

$$= \ln\left(\frac{1+e}{1+e^{-1}}\right)$$

$$= \ln\left(\frac{1+e}{\frac{1+e}{e}}\right)$$

$$= \ln e$$

$$= 1$$

$$d) \quad 8x^3 - 6x = -1$$

$$2(4x^3 - 3x) = -1 \quad \text{let } x = \cos \theta$$

$$2(4\cos^3 \theta - 3\cos \theta) = -1$$

$$\cos 3\theta = -\frac{1}{2}$$

$$3\theta = 120^\circ, 240^\circ, 480^\circ$$

$$\theta = 40^\circ, 80^\circ, 160^\circ$$

$$\therefore x = \cos 40^\circ, \cos 80^\circ, \cos 160^\circ$$

$$= 0.766, 0.174, -0.940$$

Q7

a)

$$\int \frac{dx}{x^2 + 6x - 7}$$
$$= \int \frac{dx}{(x+7)(x-1)} \quad \frac{1}{(x+7)(x-1)} = \frac{A}{x+7} + \frac{B}{x-1}$$

$$\therefore 1 = A(x-1) + B(x+7)$$

$$x=1: 1 = 8B$$

$$B = \frac{1}{8}$$

$$x=-7: 1 = -8A$$

$$A = -\frac{1}{8}$$

$$= \int \frac{-\frac{1}{8}}{x+7} + \frac{\frac{1}{8}}{x-1} dx$$

$$= -\frac{1}{8} \ln(x+7) + \frac{1}{8} \ln(x-1)$$

$$= \frac{1}{8} \ln\left(\frac{x-1}{x+7}\right) + C$$

b) $\int_0^{\frac{3}{\sqrt{2}}} \sqrt{9-x^2} dx$

$$x = 3 \sin \theta$$

$$dx = 3 \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} \sqrt{9-9\sin^2\theta} \cdot 3\cos\theta d\theta$$

$$= 9 \int_0^{\frac{\pi}{4}} \sqrt{1-\sin^2\theta} \cdot \cos\theta d\theta$$

$$= 9 \int_0^{\frac{\pi}{4}} \cos^2\theta d\theta$$

$$= \frac{9}{2} \int_0^{\frac{\pi}{4}} (1 + \cos 2\theta) d\theta$$

$$= \frac{9}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{4}}$$

$$= \frac{9}{2} \left(\frac{\pi}{4} + \frac{1}{2} \right)$$

c) i) $(x-k)^2$ is a factor $\Rightarrow k$ is a double root

$$\begin{aligned}\therefore p'(x) &= 3x^2 - 10x + 8 \\ &= (3x-4)(x-2)\end{aligned}$$

$\therefore$ double root is $\frac{4}{3}$ or 2

$$\therefore k = \frac{4}{3} \text{ or } 2$$

ii) when $k=2$

$$\begin{aligned}2^3 - 5(2)^2 + 8(2) + b &= 0 \\ b &= -4\end{aligned}$$

$$\begin{aligned}\therefore \text{Sum of roots} \quad 2 + 2 + \alpha &= 5 \\ \alpha &= 1\end{aligned}$$

$$\therefore p(x) = (x-2)^2(x-1)$$

Q 8

$$a) \int_0^{\frac{\pi}{2}} \frac{dx}{5 + 3 \cos x - 4 \sin x}$$

$$= \int_0^1 \frac{\frac{2 dt}{1+t^2}}{5 + 3\left(\frac{1-t^2}{1+t^2}\right) - 4\left(\frac{2t}{1+t^2}\right)}$$

$$= \int_0^1 \frac{2 dt}{5(1+t^2) + 3(1-t^2) - 4(2t)}$$

$$= \int_0^1 \frac{2 dt}{8 - 8t + 2t^2}$$

$$= \int_0^1 (4-t)^{-2} dt$$

$$= \left[-(t-2)^{-1} \right]_0^1$$

$$= \frac{1}{2}$$

$$b) \quad i) \quad y = \frac{x}{2} \Rightarrow x = 2y$$

$\therefore P(2y) = 0$ is required polynomial equation

$$(2y)^3 + 4(2y)^2 + 3(2y) - 3 = 0$$

$$8y^3 + 16y^2 + 6y - 3 = 0$$

$$ii) \quad \alpha\beta - 1 = \frac{\alpha\beta\gamma}{\gamma} - 1 \quad \alpha\gamma - 1 = \frac{\beta}{\beta} - 1$$

$$= \frac{\beta}{\gamma} - 1 \quad \beta\gamma - 1 = \frac{3}{2} - 1$$

$$\therefore y = \frac{3}{2} - 1$$

$$\frac{3}{2} = y + 1$$

$$x = \frac{3}{y+1}$$

$\therefore P\left(\frac{3}{y+1}\right) = 0$ is required polynomial equation

$$\left(\frac{3}{y+1}\right)^3 + 4\left(\frac{3}{y+1}\right)^2 + 3\left(\frac{3}{y+1}\right) - 3 = 0$$

$$27 + 36(y+1) + 9(y+1)^2 - 3(y+1)^3 = 0$$

$$27 + 36y + 36 + 9y^2 + 18y + 9 - 3y^3 - 9y^2 - 9y - 3 = 0$$

$$69 + 45y - 3y^3 = 0$$

$$y^3 - 15y - 23 = 0$$

$$c) \quad 1) \quad I_n = \int_1^e (1 - \ln x)^n dx$$

$$u = (1 - \ln x)^n \quad v = x$$

$$u' = n(1 - \ln x)^{n-1} \left(-\frac{1}{x}\right) \quad v' = 1$$

$$\begin{aligned} \therefore I_n &= x(1 - \ln x)^n \Big|_1^e + \int_1^e n(1 - \ln x)^{n-1} \left(\frac{1}{x}\right) dx \\ &= e(1 - \ln e) - 1(1 - \ln 1) + n \int_1^e (1 - \ln x)^{n-1} dx \\ &= -1 + n I_{n-1} \end{aligned}$$

$$\begin{aligned} ii) \quad I_3 &= -1 + 3 I_2 \\ &= -1 + 3 [-1 + 2 I_1] \\ &= -4 + 6 I_1 \end{aligned}$$

$$\begin{aligned} &= -4 + 6 [-1 + I_0] \quad I_0 = \int_1^e (1 - \ln x)^0 dx \\ &= -10 + 6(e - 1) \quad = [x]_1^e \\ &= 6e - 16 \quad = e - 1 \end{aligned}$$

Q9

a) i) $\int_4^{16} \frac{\sqrt{x}}{x-1} dx$

$$x = u^2$$

$$dx = 2u du$$

$$= \int_2^4 \frac{u}{u^2-1} \cdot 2u du$$

$$= \int_2^4 \frac{2u^2 du}{u^2-1}$$

$$= 2 \int_2^4 \frac{u^2-1+1}{u^2-1} du$$

$$= 2 \int_2^4 \left(1 + \frac{1}{u^2-1} \right) du$$

$$\frac{1}{u^2-1} = \frac{A}{u-1} + \frac{B}{u+1}$$

$$\therefore 1 = A(u+1) + B(u-1)$$

$$A = \frac{1}{2}, B = -\frac{1}{2}$$

$$= 2 \int_2^4 \left(1 + \frac{\frac{1}{2}}{u-1} - \frac{\frac{1}{2}}{u+1} \right) du$$

$$= \left[2u + \ln(u-1) - \ln(u+1) \right]_2^4$$

$$= (8 + \ln 3 - \ln 5) - (4 + \ln 1 - \ln 3)$$

$$= 4 + 2\ln 3 - \ln 5$$

ii) $\int_4^{16} \frac{\ln(x-1)}{\sqrt{x}} dx$

$$u = \ln(x-1)$$

$$v = 2x^{\frac{1}{2}}$$

$$u' = \frac{1}{x-1}$$

$$v' = x^{-\frac{1}{2}}$$

$$= 2\sqrt{x} \ln(x-1) \Big|_4^{16} - 2 \int_4^{16} \frac{\sqrt{x}}{x-1} dx$$

$$= (8\ln 15 - 4\ln 3) - 2(4 + 2\ln 3 - \ln 5)$$

$$= 8\ln 15 - 8\ln 3 + 2\ln 5 - 8$$

$$= 10\ln 5 - 8$$

b) i)

$$z_1 = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$$

$$z_2 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$z_3 = -1$$

$$z_4 = \cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5} \quad \text{or} \quad \cos \frac{3\pi}{5} - i \sin \frac{3\pi}{5}$$

$$z_5 = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5} \quad \text{or} \quad \cos \frac{\pi}{5} - i \sin \frac{\pi}{5}$$

ii)

$$\alpha = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$$

$$z_2 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$= \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \right)^3$$

$$= \alpha^3$$

$$z_4 = \cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5}$$

$$= \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \right)^7$$

$$= \alpha^7$$

$$= \alpha^5 \cdot \alpha^2$$

$$= -1 \times \alpha^2$$

$$= -\alpha^2$$

$$z_5 = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5}$$

$$= \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \right)^9$$

$$= \alpha^9$$

$$= \alpha^5 \cdot \alpha^4$$

$$= -1 \times \alpha^4$$

$$= -\alpha^4$$

iii) quadratic with roots $\alpha - \alpha^4$ and $\alpha^3 - \alpha^2$

$$\begin{aligned}\text{Sum} &= \alpha - \alpha^4 + \alpha^3 - \alpha^2 \\ &= 1\end{aligned}$$

$$\begin{aligned}\text{as } -1 + \alpha + \alpha^3 - \alpha^4 - \alpha^2 &= 0 \\ \text{sum of roots of } x^5 + 1 &= 0\end{aligned}$$

$$\begin{aligned}\text{product} &= (\alpha - \alpha^4)(\alpha^3 - \alpha^2) \\ &= \alpha^4 - \alpha^3 - \alpha^7 + \alpha^6 \\ &= \alpha^4 - \alpha^3 - \alpha^5 \cdot \alpha^2 + \alpha^5 \cdot \alpha \quad (\alpha^5 = -1) \\ &= \alpha^4 - \alpha^3 + \alpha^2 - \alpha \\ &= -(\alpha - \alpha^2 + \alpha^3 - \alpha^4) \\ &= -1\end{aligned}$$

$\therefore$ required quadratic is $x^2 - (\text{sum of roots})x + \text{product} = 0$

$$x^2 - x - 1 = 0$$